

UI/UX Design Specification & Concept Document

Core Identity Requirement:
“My name is Fajwan Chanjwok.”

Application Name:	FCPortfolio
Student Full Name:	Fajwan Chanjwok (Legal: Deng Fajwan Changjwok Fajwan)
Registration Number:	25/28110 (REGDIGITS: 2528110)
Verification Code:	MOB-A2-8110
Target Package:	rw.ac.ines.ug2528110.fcportfolio
Platform & Framework:	Android Standalone Release (React Native, Expo SDK 51, TypeScript)

1. Target Audience & Recruiter Persona

The primary user persona is an engineering recruiter, technical hiring manager, or innovation challenge judge. These evaluators spend an average of 90 to 120 seconds inspecting a portfolio app. They require immediate proof of identity, clear verification credentials, clean responsive navigation, evidence of practical coding capability, and 100% offline reliability during on-site interviews without dependency on venue Wi-Fi or Expo Go.

2. Visual Direction & Design Philosophy

Professional Minimalism: Designed with an uncluttered dark-slate and vibrant-sky blue aesthetic (#0F172A and #0284C7), evoking high technical rigor. All typography conforms to WCAG AA contrast standards (>7:1 for headers, >4.5:1 for body text).

Content-First Density: Cards present scannable metadata, distinct tag badges, and clear touch targets (minimum 48x48 dp).

Offline Transparency: Every screen features offline indicators to reassure recruiters that the application is fully standalone.

Page 2: User Flow & Information Architecture

1. Information Architecture (IA) Hierarchy

Level 1: Root Tabs	Level 2: Screen View	Level 3: Interactivity / Stack Flow
Tab 1: Home	HomeScreen (Hero Card, Exact Statement, Stats)	Quick navigation triggers to Projects, Skills, Profile.
Tab 2: Skills	SkillsScreen (Segmented: Skills & Education)	Interactive tab toggle between 5 skills and academic timeline.
Tab 3: Projects (Stack)	ProjectsListScreen (Project overview cards)	Navigates via Native Stack to ProjectDetailsScreen with back state preservation.
Tab 4: Profile	ProfileEditorScreen (Controlled form & media)	Real-time validation, Camera/Gallery launch, AsyncStorage save & reset.
Tab 5: About	AboutScreen (Verification card & Release info)	Presents MOB-A2-8110, reg number, app version, and privacy statement.

2. Critical User Flows

Flow A: Recruiter Rapid Review (60s)

Launch installed APK → Home screen displays 'My name is Fajwan Chanjwok.' and MOB-A2-8110 → Tap Projects → Select 'RP2350 Telemetry' → Inspect problem statement, hardware architecture, and student contribution → Press Android hardware or header Back button → Session preserved on Projects list.

Flow B: Profile Customization & Local Persistence Flow

Tap Profile tab → Tap 'Choose from Gallery' or 'Take Photo' → Grant permission → Image preview updates → Modify Headline and Bio → Real-time validation checks 3 rules → Tap 'Save Profile Changes' → AsyncStorage saves JSON → Kill app process → Re-launch APK → Profile data and image remain restored.

Flow C: Offline / Airplane Mode Verification Flow

Enable Airplane Mode → Open app → Offline banner displays green status → All screens, projects, and skills fully accessible → External GitHub repo link displays '(Requires Internet)' tag.

Wireframe 1: Home Screen (Introduction & Core Statement)

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█ [.] 100% Offline Capable • Airplane Mode Verified █ ← Offline Status Banner
██████████████████████████████████████████████████████████████████████████
█ [Avatar: 84x84] [MOB-A2-8110] [Available Pill] █ ← Verification & Status
█ █
█ "My name is Fajwan Chanjwok." █ ← Mandatory Core Sentence
█ Software Engineering Student & Embedded Developer █ ← Headline
█ Undergraduate SWE student at INES-Ruhengeri... █ ← Bio Overview
█ Primary Focus: [ React Native & TypeScript ] █
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█ [5+ Skills] [2 Projects] [100% Offline] █ ← Metric Highlights
██████████████████████████████████████████████████████████████████████████
█ [ > Featured Projects: RP2350 IoT & SWE 3513 AI ] █ ← Direct Nav Touch Target
█ [ > Skills & Academic Training Pathway ] █
█ [ > Edit Profile & Avatar Media ] █
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[Tab Bar: Home (Active) | Skills | Projects | Profile | About]

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[illegible]

Page 4: Annotated Wireframes — Projects & Profile Editor

Wireframe 3: Project Detail Stack Route

■ [← Back to Projects] ■ ← *Stack Back Navigation*
■ **RP2350 Dual-Core Sensor Gateway & Telemetry Unit** ■
■ Embedded IoT monitoring node with low-power telemetry ■
■ [RP2350] [MicroPython] [UART/I2C] [Power Optimization]■
■ **The Problem:** ■
■ Agricultural field stations operate without cellular...■
■ **Student Contribution:** (Highlighted Card) ■
■ Designed core firmware on Waveshare RP2350. Configured■
■ dual-core partitioning (Core 0 ADC, Core 1 Flash)... ■
■ [!] Optional External Repo (Requires Internet) ■ ← *Offline Disambiguation*
■ [View Source on GitHub →] ■

Wireframe 4: Profile Editor (Controlled Inputs & Validation States)

■ [Avatar: 110x110 Circle Preview] ■ ← *Live Image Preview*
■ [Choose from Gallery] [Take Photo] [Remove] ■ ← *Media Action Controls*
■ Professional Headline * ■
■ [Dev] ■ ← *Invalid Input State*
■ **✗ Headline must be at least 5 characters long.** ■ ← *Field-Level Error*
■ Primary Technical Skill * ■
■ [React Native & TypeScript [X]]■
■ Short Biography * ■
■ [Undergrad student... (120/25 chars min)]■
■ [Save Profile Changes (DISABLED)] ■ ← *Blocked Invalid Save*
■ [Reset to Default Profile] ■

Page 5: Visual & Accessibility Engineering Decisions

1. Color Palette & WCAG AA Contrast Compliance

Token Name	Hex Value	Contrast on White/Surface	Functional Usage
Primary	#0284C7	4.68:1 (Passes AA)	Primary buttons, active tabs, brand accents.
Primary Dark	#0369A1	7.12:1 (Passes AAA)	Section titles, verification code badge, headers.
Text Primary	#0F172A	16.1:1 (Passes AAA)	Body text, card titles, core statement.
Text Secondary	#475569	7.25:1 (Passes AAA)	Biographies, problem descriptions, metadata.
Danger (Error)	#DC2626	4.92:1 (Passes AA)	Field-level validation error messages, remove photo.
Success	#059669	4.55:1 (Passes AA)	Offline ready pill, active availability badge.

2. Typography & Large-Text Scalability

The typography system utilizes Android system font stacks (Roboto) with explicit relative line heights. When users increase system text size under Android Settings → Display → Font Size, the layouts employ Flexbox wrap (flexWrap: 'wrap') and ScrollView wrappers to prevent text truncation, clipping, or overlapping. Minimum touch targets are strictly maintained at 48x48 dp across all buttons and tab bar icons.

3. Accessibility Roles & Screen Reader Semantics

All custom interactive components explicitly declare `accessible={true}`, `accessibilityRole` (e.g. 'button', 'header', 'alert', 'text'), and descriptive `accessibilityLabel` strings. Field-level validation messages use `accessibilityRole='alert'` to immediately notify vision-impaired users using Android TalkBack.

Page 6: Implementation Comparison & Design Fidelity

1. Wireframe vs. Implementation Matrix

Feature / Screen	Wireframe Specification	Final Production Build Evidence	Status
Home Statement	First sentence: "My name is [name]."	Renders: "My name is Fajwan Chanjwok." with 22pt bold typography.	100% Match
Verification Code	Display MOB-A2-LLLL on About and Home.	Displayed in monospace chip: MOB-A2-8110 (derived from 25/28110).	100% Match
Navigation Stack	Bottom tabs + nested stack for Project Details.	React Navigation 5-tab bar + Native Stack with hardware back handling.	100% Match
Input Validation	At least 3 rules & blocked invalid save.	Rules for headline (5-70), bio (25-300), and skill (2-40); save button disabled.	100% Match
Media Workflow	Capture/gallery, preview, replace, remove.	Expo ImagePicker integration with camera, gallery, remove & error alerts.	100% Match
Persistence	AsyncStorage save, load, reset on restart.	JSON serialized under @fcportfolio_profile_v1; survives cold restart.	100% Match

2. Justification of Enhancements

- Enhancement 1 (Offline Transparency Banner):** Added a top-level offline status badge across the Home and Projects screens. This guarantees that during physical device evaluation, recruiters immediately recognize that the application does not rely on active network calls.
- Enhancement 2 (Clear-Text Helper in Inputs):** Added a one-tap 'Clear' button to controlled input headers to accelerate user testing during the 2-minute demonstration video.
- Enhancement 3 (Dedicated Education Timeline):** Integrated a segmented control in the Skills screen to cleanly partition technical competencies from the academic timeline, avoiding vertical scroll exhaustion.